METRIC

MIL-DTL-24694/1A(SH)
8 June 2016

SUPERSEDING
MIL-V-24694/1(SH)
28 October 1987

DETAIL SPECIFICATION SHEET

VALVE, HYDRAULIC RELIEF, 1.5 – 35 BAR, 150 LPM CARTRIDGE TYPE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24694.

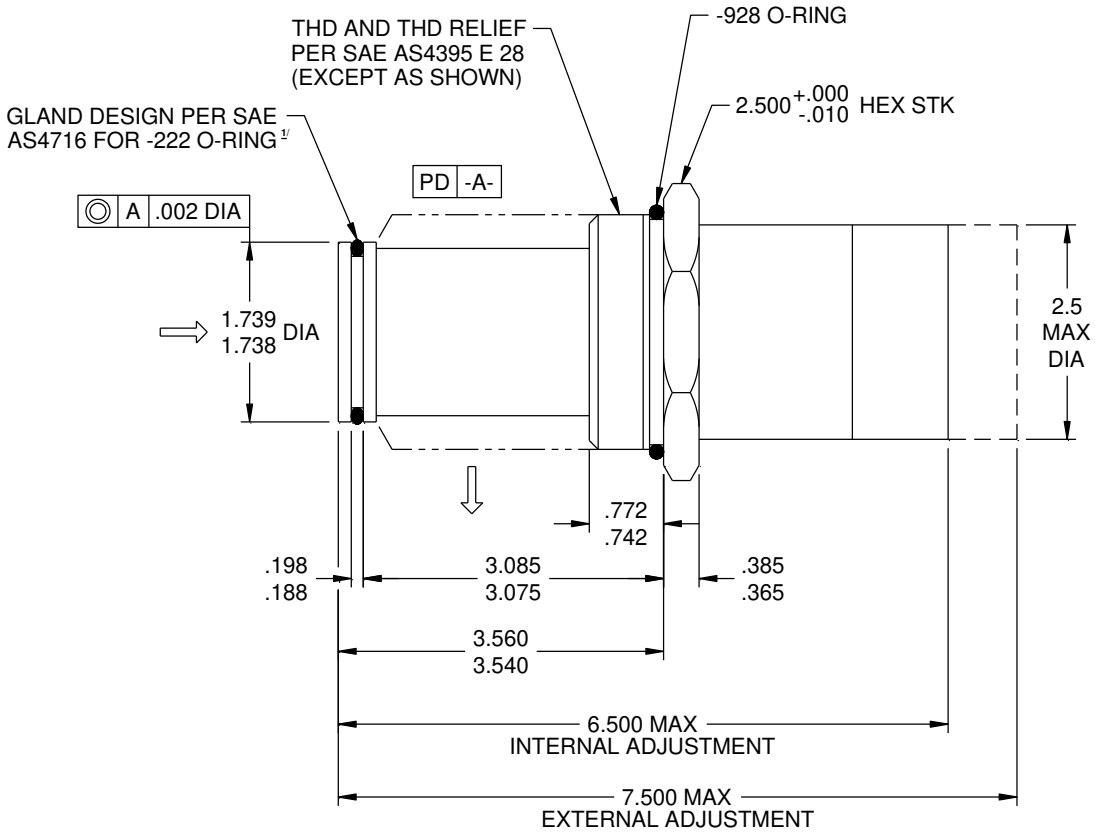


FIGURE 1. Hydraulic relief valve configuration.

FOOTNOTE:

1/ Gland dimensions of MIL-G-5514 valid for use on previously qualified designs.

NOTE:

1. All dimensions are shown in inches.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:02Z
Check the source to verify that this is the current version before use.

TABLE I. Pressure class.

Class	Nominal set differential pressure range	
	Bar (1 x 105 pascals)	lb/in^2
-A	1.4 - 2.0	20.3 - 29.0
-B	2.0 - 3.5	29.0 - 50.75
-C	3.5 - 5.0	50.75 - 72.5
-D	5.0 - 7.0	72.5 - 101.5
-E	7.0 - 10.0	101.5 - 145
-F	10.0 - 14.0	145 - 203
-G	14.0 - 20.0	203 - 290
-H	20.0 - 30.0	290 - 435
-I	30.0 - 35.0	435 - 507.5

TABLE II. Adjustment type.

Dash letters	Description
IN	Internal adjustment
EX	External adjustment

REQUIREMENTS:

1. Allowable reseat leakage 10 milliliters per minute maximum with maximum fluid viscosity of 75 centistokes.

2. Surface roughness shall not exceed 125 microinches in accordance with ASME B46.1. All burrs shall be removed and sharp edges broken at 0.005 inch minimum.

3. Material shall be in accordance with MIL-DTL-24694.

4. Rated flow shall be 150 liters per minute.

Part number: The part number shall consist of the basic number of this specification sheet, plus the applicable class letters selected from table I, and the adjustment type dash letters selected from table II. For valves whose pressure range covers more than one class, the letters designating the minimum and maximum setting are used to form the part number.

Example: M24694/1-C-IN Valve with a pressure range of 3.5 to 5 bar, internal adjustment

M24694/1-CE-IN Valve with a pressure range of 3.5 to 10 bar, internal adjustment

CHANGES FROM PREVIOUS ISSUE. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Preparing activity:
Navy – SH
(Project 4810-2016-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:02Z
Check the source to verify that this is the current version before use.